

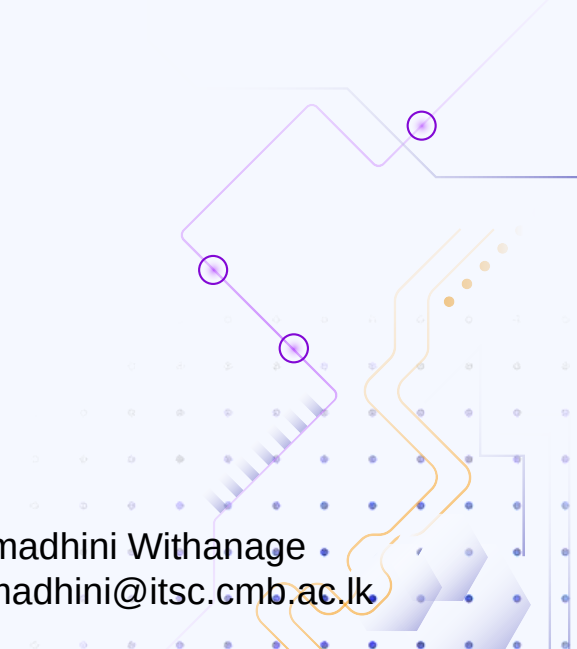


IT 3003

Advanced Programming Techniques

Java Practical Session 1

Start small, think big, and keep coding

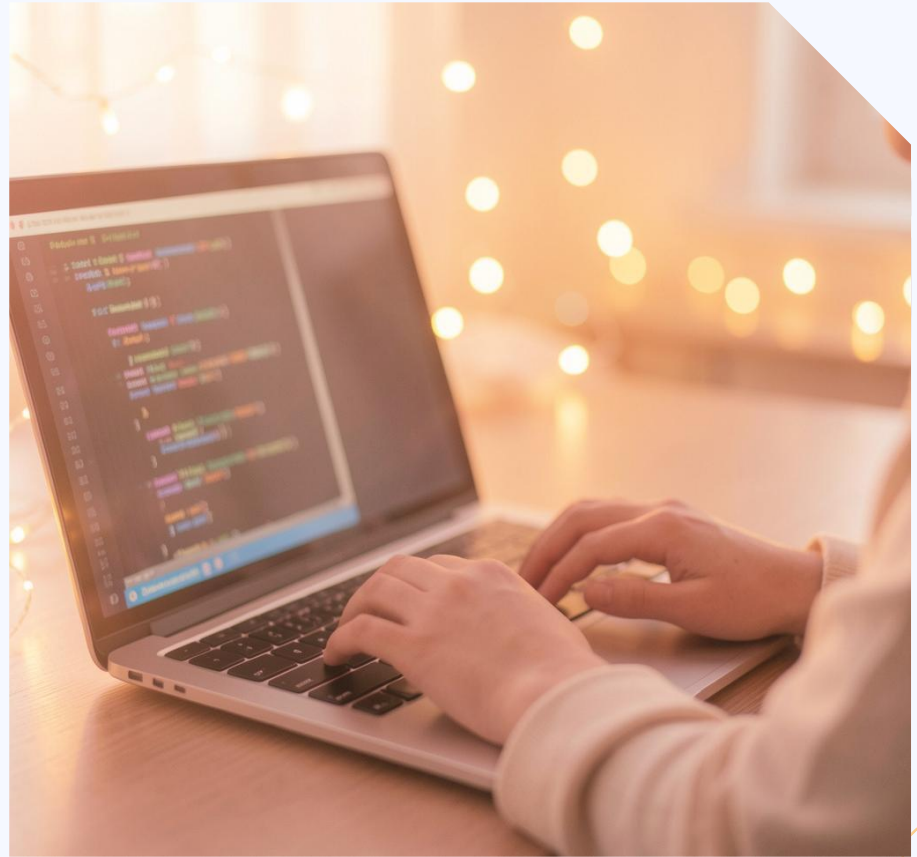


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Introduction

Programming is the art of giving clear and logical instructions to a computer to solve problems step by step.

Begin with simple concepts, focus on understanding the logic, and build gradually. Always practice, experiment, and refine your code to develop strong and confident programming skills.





Lesson Outline

- ☀️ Print Word
- ☀️ Variables & Data Types
- ☀️ Operators
- ☀️ User Input
- ☀️ Arrays
- ☀️ Control Statements - if/else
- ☀️ Control Statements – Switch Case
- ☀️ Loop – for
- ☀️ Loop – While
- ☀️ Loop – do While
- ☀️ Methods

Create First Java Project

Create a Java file named Test.java and write a program to print:

Welcome to Java Programming

```
public class Test {  
    public static void main(String[] args) {  
        System.out.println("Welcome to Java Programming");  
    }  
}
```



Compile and run the program

Compile and run the above program using Command Prompt

```
javac Test.java  
java Test
```

OUTPUT

```
Welcome to Java Programming
```

Displaying a Simple Output Message

Write a Java program to print:

Hello World

```
public class Hello {  
    public static void main(String[] args) {  
        System.out.println("Hello World");  
    }  
}
```



Displaying a Simple Output Message cont..

Modify the program to print your name and age in two separate lines.

Java source file : Hello.java

Variables & Data Types

Declare variables of type int, double, and char and print their values.

Java source file : DataTypes.java

Variables & Data Types cont..

Create a program to store a student's name, age, and GPA, then display them.

Java source file : Student.java

Operators

Write a program to perform addition, subtraction, multiplication, and division of two numbers.

Java source file : Operators.java

Operators cont..

Write a program to calculate the remainder when one number is divided by another.

Java source file : Modulus.java

User Input

Write a program to input an integer from the user and print it.

Java source file : InputExample.java

User Input cont..

Write a program to input a user's name and print

Java source file : NameInput.java

Arrays

Declare an array of 5 integers and print all elements.

Java source file : ArrayExample.java

Arrays cont..

Write a program to calculate the sum of elements in an array.

Java source file : ArraySum.java

Control Statements – if else

Write a program to check whether a number is positive or negative.

Java source file : ArraySum.java

Control Statements – if else cont..

Write a program to input marks and print grade

input value = num

num $\geq 75 \rightarrow A$

50 $\leq$ num $< 75 \rightarrow B$

else $\rightarrow C$

Java source file : Grade.java

Control Statements – Switch Case

Write a program that takes a number (1–3) and prints the corresponding day:

1 → Monday

2 → Tuesday

3 → Wednesday

Java source file : SwitchBasic.java

Control Statements – Switch Case cont..

Write a program that takes a number and performs a calculation based on user choice:

1 → Addition

2 → Subtraction

3 → Multiplication

4 → Division

Ask the user to input two numbers and display the result.

Java source file : SwitchCalculator.java

Loop – For Loop

Print numbers from 1 to 5 using a for loop.

Java source file : LoopExample.java

Loop – For Loop cont..

Write a program to calculate the sum of numbers from 1 to 10.

Java source file : LoopSum.java

Loop – WHILE LOOP

Write a program to print numbers from 1 to 5 using a while loop.

Java source file : WhileExample.java

Loop – WHILE LOOP cont..

Write a program to calculate the sum of numbers from 1 to n (user input) using a while loop.

Java source file : WhileSum.java

Loop – DO-WHILE LOOP

Write a program to print numbers from 1 to 5 using a do-while loop.

Java source file : DoWhileExample.java

Loop – DO-WHILE LOOP cont..

Write a program that keeps asking the user to enter a number until they enter 0.

Java source file : DoWhileExample.java

Methods

Create a method to print “Welcome”.

Java source file : MethodExample.java

Methods cont..

Create a method that takes two numbers and returns their sum..

Java source file : MethodSum.java

Take-Home Exercise

Lab Sheet 01



Thanks!

